NETWORKING ASSIGNMENT

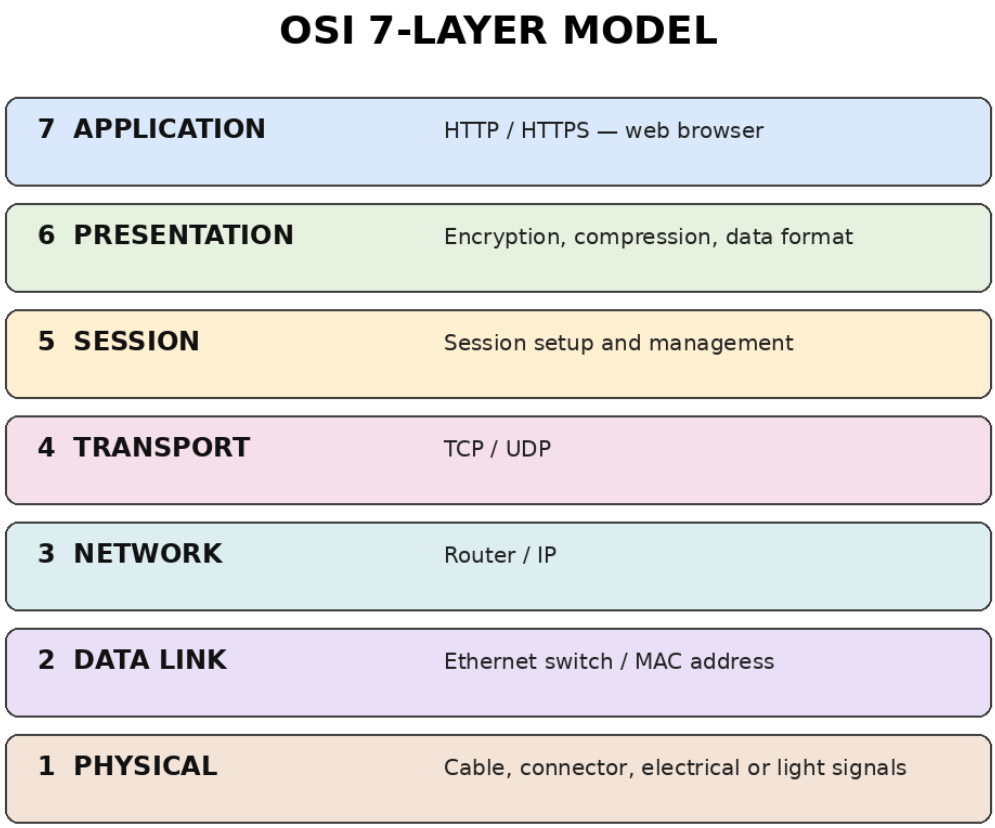
OSI 7-Layer Model and Troubleshooting

TASK 1: Draw and Label the OSI Model

QUESTION

Draw the OSI 7-layer model. Label every layer with its correct name and give one real-world device, protocol, or function associated with each layer.

ANSWER



The OSI model organizes network communication into seven layers, from physical transmission at Layer 1 to user-facing network services at Layer 7.

TASK 2: Match Scenarios to OSI Layers

QUESTION

Match each scenario to the most suitable OSI layer. Write each answer in the format: Scenario: Layer Name (Layer Number).

a) Encrypting data before sending it.
b) Routing packets across networks.
c) Converting data into electrical or light signals.
d) Opening a web page in a browser.
e) Detecting errors in Ethernet frames.

ANSWER

Scenario	Correct answer
a) Encrypting data before sending	Presentation (Layer 6)
b) Routing packets across networks	Network (Layer 3)
c) Converting data into electrical or light signals	Physical (Layer 1)
d) Opening a web page in a browser	Application (Layer 7)
e) Detecting errors in Ethernet frames	Data Link (Layer 2)

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TASK 3: OSI Layers in a Daily App

QUESTION

Choose one app you use daily and explain how at least three OSI layers help it send or receive information.

ANSWER

Example: WhatsApp

• **Application (Layer 7):** WhatsApp provides messaging and calling features and prepares the message for communication over the network.

• **Presentation (Layer 6):** Data may be encrypted and represented in a format the receiving device can process. In real systems, these functions may be handled by application and security libraries rather than a separate OSI layer.

• **Transport (Layer 4):** Transport protocols help move data between devices. TCP or UDP may be used depending on the service.

• **Network (Layer 3):** IP addresses help packets travel between the sender's device and destination servers.

• **Data Link and Physical (Layers 2 and 1):** Wi-Fi or Ethernet carries frames over the local network, while radio waves or cable signals carry the bits.

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TASK 4: Website Does Not Open, but Ping Works

QUESTION

A user cannot open a website in a browser, but pinging the website's IP address succeeds. Identify likely OSI layer(s) involved and explain how you would troubleshoot.

ANSWER

The problem is likely above basic IP connectivity, especially the Application layer (Layer 7), and may also involve the Transport layer (Layer 4) or name resolution.

1. Test the website using its IP address in the browser, if the site supports it. A successful ping only confirms that ICMP traffic reaches the host; it does not prove the web service is working.

2. Check DNS resolution. If the domain name does not resolve to the correct IP address, investigate DNS settings or the DNS server.

3. Check whether the web server is reachable on TCP port 80 (HTTP) or 443 (HTTPS). A firewall may allow ping but block web traffic.

4. Check browser proxy settings, HTTPS/TLS errors, and the website's server status.

Conclusion: IP connectivity appears to work, but DNS, TCP/HTTPS connectivity, browser configuration, or the web application may be failing.

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